A PROJECT REPORT ON

AUTODRIVE: MODERN LUXURY CAR RENTAL SYSTEM & FLEET SAAS WEB APPLICATION

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This is to certify that the project entitled "AUTODRIVE: MODERN LUXURY CAR RENTAL SYSTEM & FLEET SAAS WEB APPLICATION" has been carried out by Meet Parekh (Enrollment No: 2504070100175) under the guidance of Project Guide Name in partial fulfilment of the requirements for the 3rd Semester Minor Project (CAM6053C) of Master of Computer Application (MCA) at Silver Oak College of Computer Application, Silver Oak University, Ahmedabad during the academic year 2026-2027.

The project work embodies the original work of the candidate and has been completed to the satisfaction of the department in accordance with the university curriculum regulations.

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STUDENT DECLARATION

I hereby declare that the project entitled "AUTODRIVE: MODERN LUXURY CAR RENTAL SYSTEM & FLEET SAAS WEB APPLICATION" submitted in partial fulfilment of the requirements for the 3rd Semester of Master of Computer Application (MCA) is an original piece of work carried out by me under the guidance and supervision of Mr. Dhruv Patel, Assistant Professor, Department of Computer Application, Silver Oak University.

I further declare that this project report, or any part thereof, has not been submitted by me or any other candidate for the award of any degree, diploma, fellowship, or other similar title to this or any other university or academic institution.

All sources of literature, libraries, frameworks, open-source utilities, and technical citations used during the system design, development, and testing have been duly acknowledged in the references section.

Place: Ahmedabad
Date: _____

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ABSTRACT

The automotive rental sector is transitioning rapidly from manual, counter-based leasing operations to agile, automated digital fleet platforms. Traditional car rental systems frequently suffer from critical operational vulnerabilities, including double-booking collisions, lack of real-time slot conflict prevention, opaque pricing calculations, manual chauffeur dispatch errors, and friction in handling digital payment transactions.

This project presents the design, architecture, and full-stack implementation of "AutoDrive", a cutting-edge Luxury Car Rental System and Fleet SaaS Web Application built on Python and the Django Model-View-Template (MVT) framework. AutoDrive introduces an algorithmic mathematical Date Range Overlap Collision Detection Engine that guarantees zero double-booking conflicts across both fleet vehicles and certified executive chauffeurs. The system features dynamic tiered rental pricing (providing automated 10% and 25% discounts for multi-day rentals), multi-facet fleet discovery filters, and a seamless dual-mode payment gateway integrating the Razorpay Standard Checkout SDK alongside an in-browser 3D Secure Bank OTP Simulator for realistic offline testing.

The application incorporates comprehensive administrative business intelligence dashboards, real-time fleet telematics summaries, vehicle comparison matrices, verified customer driver reviews, and customer self-service portals. Rigorous automated verification across 27 unit and integration test suites confirmed a 100% test pass rate with zero regression. AutoDrive successfully delivers high reliability, robust data integrity, and a premium cyberpunk-luxury aesthetic tailored for contemporary automotive mobility enterprises.

Keywords: Car Rental System (CRS), Fleet Management SaaS, Django Framework, Date Range Collision Prevention, Razorpay Payment Gateway, Bank Simulator, Role-Based Access Control (RBAC), SQLite/PostgreSQL, SDLC.

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